VII SEMESTER	L	T	P	C
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20AM7S01 - RESTful API Design with Node, Express, and MongoDB				

COURSE OUTCOMES:
Upon the completion of the course students will be able to:-
CO1: Understanding RESTful API Development: Gain a solid understanding of building RESTful APIs, including key concepts, principles, and best practices.
CO2: Secure Authentication System: Learn how to implement secure user authentication using JWT and the Passport authentication library, ensuring the security of web applications.
CO3: Building Real-World Features: Implement advanced functionalities such as user following, pagination, rate limiting, and caching to create feature-rich applications.
CO4: Testing and Debugging APIs: Develop proficiency in using tools like Postman to test and debug API endpoints, ensuring error handling, input validation, and overall API reliability.
CO5: Deployment and Production Readiness: Acquire insights into deploying APIs for public access, including knowledge of cloud providers, database providers, and production optimization for performance and security.

Week 1:
- Exercise 1: Course Overview
- Exercise 2: Setting Up the Development Environment
- Exercise 3: RESTful API Concepts and Principles

Week 2:
- Exercise 4: Express App Boilerplate
- Exercise 5: Integrate Error Handling
- Exercise 6: Diving into Validations

Week 3:
- Exercise 7: Consume and Test with Postman
- Exercise 8: CRUD Operations to Create API

Week 4:
- Exercise 9: Creating the Post Model
- Exercise 10: Store Post and Multer
- Exercise 11: Adding Post Index and Show Routes

Week 5:
- Exercise 12: Update Post Route
- Exercise 13: Delete Post Route
- Exercise 14: JWT and User Model

Week 6:
- Exercise 15: Passport for Authentication
- Exercise 16: Login Endpoint
- Exercise 17: Signup Endpoint

Week 7:
- Exercise 18: Authenticate Routes
- Exercise 19: Following Feature Considerations
- Exercise 20: User Schema Tweaks

Week 8:
- Exercise 21: Follow Endpoint
- Exercise 22: Fix Posts Index Route
- Exercise 23: Consume and Test New Features

Week 9:

- Exercise 24: Add Posts Pagination
- Exercise 25: Integrate Rate Limiting
- Exercise 26: Discover Caching and Redis

Week 10:

- Exercise 27: How to Cache Queries?
- Exercise 28: Serve Valid Data
- Exercise 29: Explore the Cloud Provider

Week 11:

- Exercise 30: Explore the Database Provider
- Exercise 31: Production Tweaks to Codebase
- Exercise 32: Final Testing of Public API

Week 12:

- Exercise 33: Wrap-Up

Course source link:
https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0133014041907363841706 6_shared/overview